Log-Normal Distribution in Lifetime Modelling: Definition, Applications, and Comparison with Weibull Models

Kodanda Challa, 25727

1 Introduction

A Log-Normal (LN) distribution arises when the natural logarithm of a positive random variable follows a Gaussian distribution, a structure commonly encountered in lifetime modelling. In the context of reliability engineering and statistical data analysis, this distribution is paramount because it models phenomena where the accumulation of effects is multiplicative rather than additive.

The theoretical justification comes from applying the Central Limit Theorem to multiplicative random mechanisms, where the logarithm of accumulated effects converges toward normality.

If a random variable X follows a Normal distribution such that

$$X \sim N(\mu, \sigma^2)$$

then the variable Y follows a Log-Normal distribution

$$Y = \exp(X) = LN(\mu, \sigma^2)$$

with mean $\mu \in (-\infty, +\infty)$ and standard deviation $\sigma > 0$ (logarithm scale), Y is the set of positive real number ($y > 0$).
The probability density function (PDF) is mathematically defined as:

$$f_Y(y) = \frac{1}{y\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln y - \mu)^2}{2\sigma^2}\right) \quad (1)$$
$$y > 0$$

The cumulative distribution function (CDF) is defined as:

$$F_Y(y) = \Phi\left(\frac{\ln y - \mu}{\sigma}\right)$$

Mean (Expected Value) of Y: The mean of a Log-Normal distribution is shifted by the variance.

$$E(Y) = \exp\left(\mu + \frac{\sigma^2}{2}\right) \quad (2)$$

In a normal distribution, the mean and median are the same (μ). However, because the exponential function e^x is convex (term $\sigma^2/2$), the higher values in the

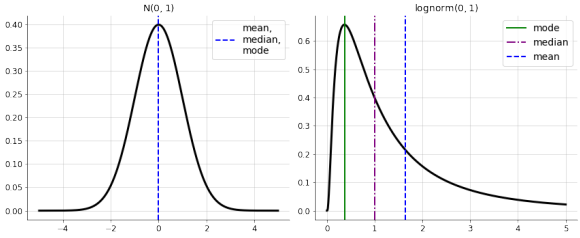


Figure 1: Normal vs Log-Normal distribution

Table 1: Properties of the Log-Normal Distribution

Property	**Formula**
Relationship	$Y = e^X,\ X \sim N(\mu, \sigma^2)$
Median	e^{μ}
Mode (Peak)	$e^{\mu-\sigma^2}$
Mean	$e^{\mu+\sigma^2/2}$
Variance	$(e^{\sigma^2}-1)e^{2\mu+\sigma^2}$

right tail of the normal distribution are "stretched" more than the lower values are "compressed"(right-skewed). This pulls the mean to the right of the median (Figure 1).

Variance of Y: The variance measures the spread of the failure times. It is expressed as

$$V(Y) = \exp(2\mu + 2\sigma^2) - \exp(2\mu + \sigma^2) \quad (3)$$

$$V(Y) = (\exp(\sigma^2) - 1)\exp(2\mu + \sigma^2) \quad (4)$$

As σ increases, the variance of Y increases exponentially. This is why the Log-Normal distribution is often used in electronics and fatigue; small variations in the degradation rate (σ) lead to very large spreads in the actual time-to-failure. Refer to Table 1 for Log Normal distribution properties.

This report focuses not only on the mathematical formulation of the Log-Normal distribution but also on the conditions under which its physical assumptions justify its use over alternative lifetime models, particularly the Weibull distribution.

2 Fields of applications

The Log-Normal Distribution is a fundamental statistical tool in reliability engineering because it models processes where change is multiplicative rather than additive. In systems prone to fatigue and degradation, damage often evolves as a percentage of previous damage, making this distribution more physically representative than the Normal (Gaussian) distribution. It is uniquely suited for modeling processes where the outcome is the result of many small independent multiplicative factors, rather than additive ones.

Primary Fields of Application

1. **Reliability Engineering and Fatigue Life:** In mechanical engineering, the log-normal distribution is used to model *Fatigue Life* , specifically when failure is caused by the growth of a dominant crack.

    - **Multiplicative Degradation:** Since each increment of crack growth is proportional to the current crack length, the total damage follows a multiplicative process.[1]
    - **Electronics:** It models the *Electromigration* in microchips, where the time-to-failure (TTF) depends on the cumulative effect of metallic ion displacement.[2]

2. **Geosciences and Material Science**

   - **Hydrology:** Modeling the distribution of extreme rainfall and river discharge volumes.[3]
   - **Particle Size:** The distribution of mineral grains in crushed ore or the size of aerosol particles in atmospheric research is typically log-normal due to the repeated fracturing process (multiplicative breakage).[4,5]

3. **Economics and Bio-Statistics**

   - **Income Distribution:** It is used to represent the distribution of wealth in a population where *wealth begets wealth.*[6,7]
   - **Pharmacokinetics:** Modeling the concentration of a drug in the bloodstream over time, where the rate of elimination is proportional to the current concentration.[8]

2.1 An example of application

In mechanical engineering, *the Log-normal distribution* is the *gold standard* for modeling fatigue life because degradation is rarely additive; it is almost always multiplicative.

If a crack grows by a certain percentage each cycle, the total crack length is the product of many small growth factors. By the *Central Limit Theorem*, the sum of the *logarithms* of these factors tends toward a normal distribution, meaning the crack length itself follows a log-normal distribution.

Calculation of Fatigue Life

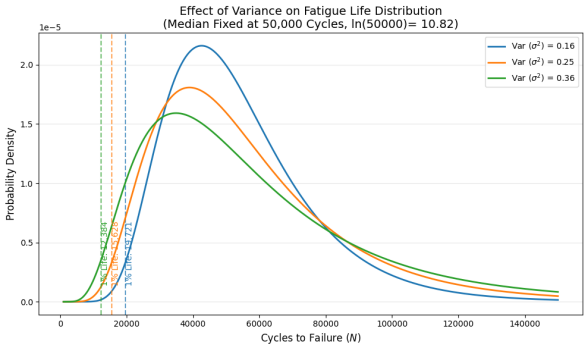


Figure 2: Log-Normal distribution for different value of variance[9]

Problem: Consider the fatigue life Y (measured in cycles) of wing spar of an aircraft, where failure is caused by the growth of a dominant crack. Since crack growth is proportional to current crack length, the process is multiplicative and follows a Log-Normal distribution.What is the minimum number of cycles y such that the probability of failure is 1% ($P(Y \leq y) = 0.01$), given that the log-scale mean μ is 10.82 and the log-scale variance σ^2 is 0.16?

Solution: We model $Y = e^X$, so $Y \sim LN(\mu = 10.82, \sigma^2 = 0.16)$. The corresponding normal variable is $X \sim N(10.82, 0.16)$.

$$P(Y \leq y) = 0.01$$

$$P(X \leq \ln y) = 0.01$$

$$P(Z \leq \frac{\ln y - 10.82}{\sqrt{0.16}}) = 0.01$$

$$\frac{\ln y - 10.82}{0.4} = -2.326$$

$$\ln y = 9.8896$$

$$y = \exp(9.8896) \approx 19,724 \; cycles$$

Conclusion: The minimum number of cycles to ensure that the probability of failure is no more than 1% is approximately 19,724 cycles.

When manufacturing quality drops (higher variance), the average life might not change much, but the Design Life (where it is safe to fly the aircraft) drops drastically. In our example, increasing variance from 0.16 to 0.36 drops the safe limit from $\approx 19,700$ cycles to roughly $\approx$ 12,000 cycles as shown in the Figure 2.

3 Log-Normal distribution in comparison to a Weibull distribution

The comparison between *Log-Normal* and *Weibull* lifetime models directly connects to core course topics, including transformation of random variables, cumulative distribution functions, hazard rate analysis, and model selection based on physical assumptions.

In particular, the choice between *multiplicative* and *weakest-link* representations reflects the broader statistical principle that model structure must align with the underlying data-generating mechanism

3.1 Transformations of Random Variables

The Log-normal distribution is defined by the exponential transformation of a normal random variable.

$$Y = e^X, X \sim N(\mu, \sigma^2)$$

Then the transformed variable follows a Log-normal distribution. Because this is a monotonic transformation, the properties of Y (CDF, PDF, and quantiles) are derived directly from the Normal distribution using the change-of-variables technique.

The Weibull distribution shares a similar "transformative" relationship, but its parent is the Exponential distribution. If we have:

$Y = \beta^{\frac{\alpha-1}{\alpha}} X^{1/\alpha}, \quad X \sim \text{Exp}\left(\frac{1}{\beta}\right)$

The transformed variable then follows a Weibull distribution, denoted as

$$Y = weibull(\alpha, \beta)$$

Refer Table 2 for Weibull parameters (Mean, Variance, CDF, PDF).

3.2 Hazard Rates and Failure-Mechanism Interpretation

Log-Normal Distribution: It has a unique *upside-down U* hazard rate. The probability of failure initially increases, reaches a peak, and then decreases over time.

Weibull Distribution: This distribution is often derived from powers of an exponential random variable. In reliability analysis, the Weibull distribution is valued for its flexibility, as its shape parameter (α) enables representation of multiple failure-rate behaviors. allows it to model decreasing ($\alpha < 1$), constant ($\alpha = 1$) (Exponential distribution), or increasing ($\alpha > 1$) failure rates.
Increasing α while keeping the scale (β) constant results in Symmetry Evolution.

The distribution shifts from being right-skewed to nearly symmetric. At $\alpha \approx 3.6$, it approximates a Normal distribution, mirroring the bell shape often seen in Log-Normal models with low variance (Figure 3).

Table 2: Properties of the Weibull Distribution

Property	**Formula**
Relationship	$Y = \beta^{\frac{\alpha-1}{\alpha}} X^{1/\alpha}, \alpha > 0, \quad X \sim \text{Exp}\left(\frac{1}{\beta}\right)$
CDF	$\text{f}_Y(y) = \frac{\alpha}{\beta^\alpha} y^{\alpha-1} e^{-(y/\beta)^\alpha}, \quad y > 0$ $\text{P(Y} \leq y) = 1 - e^{-(y/\beta)^\alpha}$
Mean	$E(Y) = \beta\,\Gamma\left(1 + \frac{1}{\alpha}\right)$
Variance	$\text{Var}(Y) = \beta^2 \left[\Gamma\left(1 + \frac{2}{\alpha}\right) - \left[\Gamma\left(1 + \frac{1}{\alpha}\right)\right]^2\right]$

3.3 *Numerical Comparison Problem*

Problem: An engineering team is assessing the reliability of a high-performance turbine blade. They need to calculate the probability that a blade will survive at least *800 hours*.
Sample Dataset ($n = 5$ blades) (300, 500, 665, 900, 1,500) hours

Model A (Log-Normal):
Solution:

$$\ln(300) \approx 5.70$$
$$\ln(500) \approx 6.21$$
$$\ln(665) \approx 6.50$$
$$\ln(900) \approx 6.80$$
$$\ln(1500) \approx 7.31$$

The parameters are $\mu = 6.51$ and $\sigma = 0.61$

$$P(Y > 800) = 1 - P(Y \leq 800)$$

Transform to Standard Normal (Z):

$$P(X \leq \ln(800))$$

where $X \sim N(6.51, 0.61^2)$.
Standardize:

$$Z = \frac{\ln(800) - 6.51}{0.61} \approx \frac{6.68 - 6.51}{0.61} = 0.28$$

Lookup Probability: From a Standard Normal table

$$\Phi(0.28) \approx 0.6103$$

Final Result:

$$P(Y > 800) = 1 - 0.6103 = \mathbf{0.3897} = 38.97\%$$

Hypothesis Testing: Manufacturer's claim that the log-scale mean life is $\mu_0 = 7.0$.

Null Hypothesis (H_0): $\mu = 7.0$

Alternative Hypothesis (H_1): $\mu < 7.0$

sample mean $\bar{x} = 6.51$ and sample standard deviation $s = 0.61$ from our dataset
T-statistic is calculated as:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{6.51 - 7.0}{0.61/\sqrt{5}} \approx -1.796$$

With $n - 1 = 4$ degrees of freedom, the critical value t_{crit} at $\alpha = 0.05$ is approximately -2.132. Since our calculated t (-1.796) is not less than -2.132, we cannot reject H_0 with this specific significance level (α).

Model B (Weibull):
Solution:
Transforming the non-linear CDF into a linear equation, estimate the distribution's

parameters (α and β) using simple linear regression.
The Transformation Process Starting with the Weibull CDF:

$$F(t) = 1 - e^{-(t/\beta)^{\alpha}}$$

Rearrange the equation to

$$1 - F(t) = e^{-(t/\beta)^{\alpha}}$$

Taking ln of both sides yields

$$\ln(1 - F(t)) = -(t/\beta)^{\alpha}$$

Multiply by -1 to get

$$-\ln(1 - F(t)) = (t/\beta)^{\alpha}$$

Taking ln again results in

$$\ln(-\ln(1 - F(t))) = \alpha \ln(t/\beta)$$

Using log properties, the final linear form is:

$$\ln(-\ln(1 - F(t))) = \alpha \ln(t) - \alpha \ln(\beta)$$

Mapping to a Straight Line

$$(y = mx + c)$$

Vertical Axis (y): probability scale

$$\ln(-\ln(1 - F(t)))$$

Horizontal Axis (x): time scale

$$\ln(t)$$

Slope (m): α (the shape parameter).
Intercept (c): $-\alpha \ln(\beta)$ (scale parameter)

Shape parameter $\alpha = 1.7$ and Scale parameter $\beta = 894.3$ hours. [α, β calculation[9]]

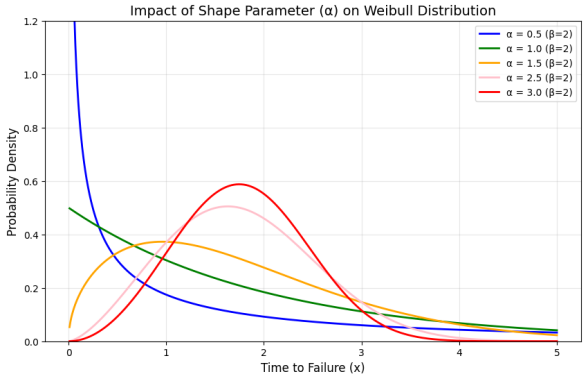


Figure 3: Weibull distribution for different values of shape parameter (α), keeping scale factor (β) constant[9]

Direct CDF Calculation:

$$P(Y \leq 800) = 1 - e^{-(800/894.3)^{1.70}}$$

$$1 - e^{-(0.895)^{1.7}} = 1 - e^{-0.828}$$

Survival Probability:

$$P(Y > 800) = e^{-0.828} \approx \mathbf{0.4369} = 43.69\%$$

Comparison This numerical example illustrates that model selection substantially alters the inferred reliability level for identical operating thresholds. (38.97% vs 43.69%).

In this case, since the Weibull model assumes that the failure rate increases over time, which is consistent with the mechanical wear of a turbine blade. The logarithmic Normal model might be less appropriate if the failure rate is expected to continue increasing indefinitely as the blade ages.

3.4 *Advantages and Limitations Comparison*

1. Failure Rate Behavior
Weibull Advantage: Since $\alpha = 1.7$ (>

1) in our problem, the Weibull model assumes that the failure rate increases over time, which perfectly matches the mechanical wear-out.
Log-Normal Limitation: Although the Log-Normal hazard rate can decrease asymptotically, this does not imply physical 'immortality', but rather reflects a limitation of the model when extrapolated beyond the domain supported by empirical data. In practice, this highlights the importance of model validation within the observed lifespan range. which is often physically incorrect for bearings.
2. Application Scope
Log-Normal Advantage: It is mathematically superior when the data is the result of multiplicative factors (like growth or interest) rather than just "time to failure".
Weibull Limitation: While excellent for the "weakest link" in a mechanical system, it is less effective at modeling biological or economic measurements where traits like height or income are driven by compound growth. Refer to Table 3 for a comparison.

Feature	**Log-Normal Distribution**	**Weibull Distribution**
Primary Strength	Modeling "wear-out" in devices where changes are proportional to the current state.	High flexibility; can quantify product reliability across different life-cycle stages.
Failure Rate	Failure rate typically increases then decreases over time, which can be a limitation for purely degrading systems.	Can represent different failure behaviors: decreasing ($\alpha < 1$), constant ($\alpha = 1$), or increasing ($\alpha > 1$).
Physical Basis	Based on the Central Limit Theorem applied to products of random variables.	Based on Extreme Value Theory (modeled on the "weakest link" principle).

Table 3: Log-Normal vs Weibull Distribution Comparison

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